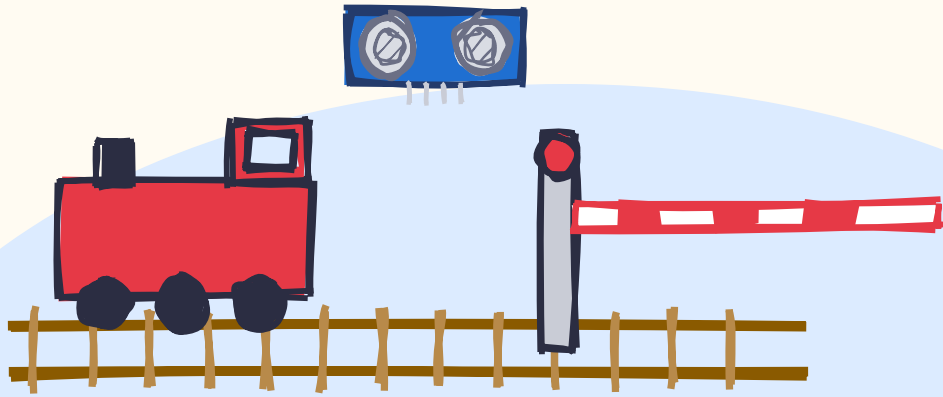
Little Explorers · Class 10

Automatic Railway Crossing Gate

Gate down, bell rings: the train is coming!

Distance sensor | Servo gate | Safety bell



Did you know?
Pakistan has thousands of railway crossings. Many have gates and bells to keep cars and people safe.

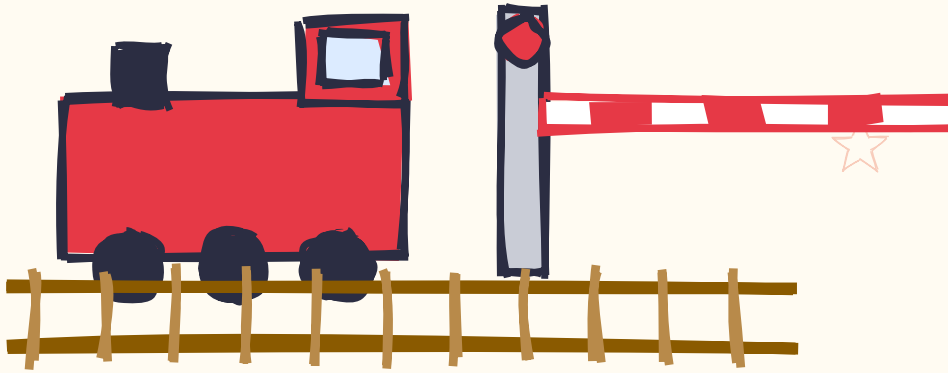
LEARN

Let's wonder...

How does the gate know a train is coming?

Ding ding ding!

At a railway crossing the red lights flash, a bell rings and the gate comes down **before** the train arrives. Nobody is standing there pressing a button. How does it know?



A guard?

It hears it?

A sensor!

Today: we build a **model crossing** that sees the train and lowers the gate!

Learning objectives

Today I can...

1	**See:** explain how the sensor 'sees' the train with **echoes**	**Proof:** clap and hear an echo, then explain it
2	**Move:** make the **gate** go down when the train is close	**Proof:** our gate comes down for the toy train
3	**Safety:** explain why crossings keep people **safe**	**Proof:** tell a friend one railway safety rule

Objective 3 · Safety · video

Video time!



Railroad Crossing Gates and How They Work

SouthernRailFan · about 6 minutes · youtu.be/KS9Om_-UqRs

Watch for...

1. What happens **before** the train arrives?
2. What colour are the flashing lights?
3. When does the gate go back **up**?

Scan to watch

After the video: Act it out! One child is the gate arm, one is the bell. Your partner is the train. Go!

Objective 1 · See

The sensor uses echoes

The idea

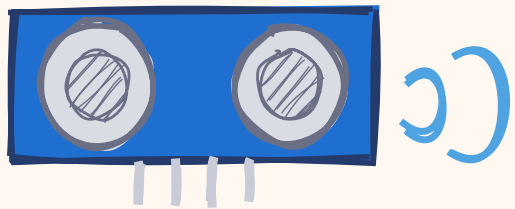
Our distance sensor has two round 'eyes'. One **shouts** a sound too high for us to hear. The other **listens** for the echo.

A **quick** echo means the train is **close**. A slow echo means it is far away.

Try it

Clap in a big room. Can you hear an echo?

Example: shouting in a cave



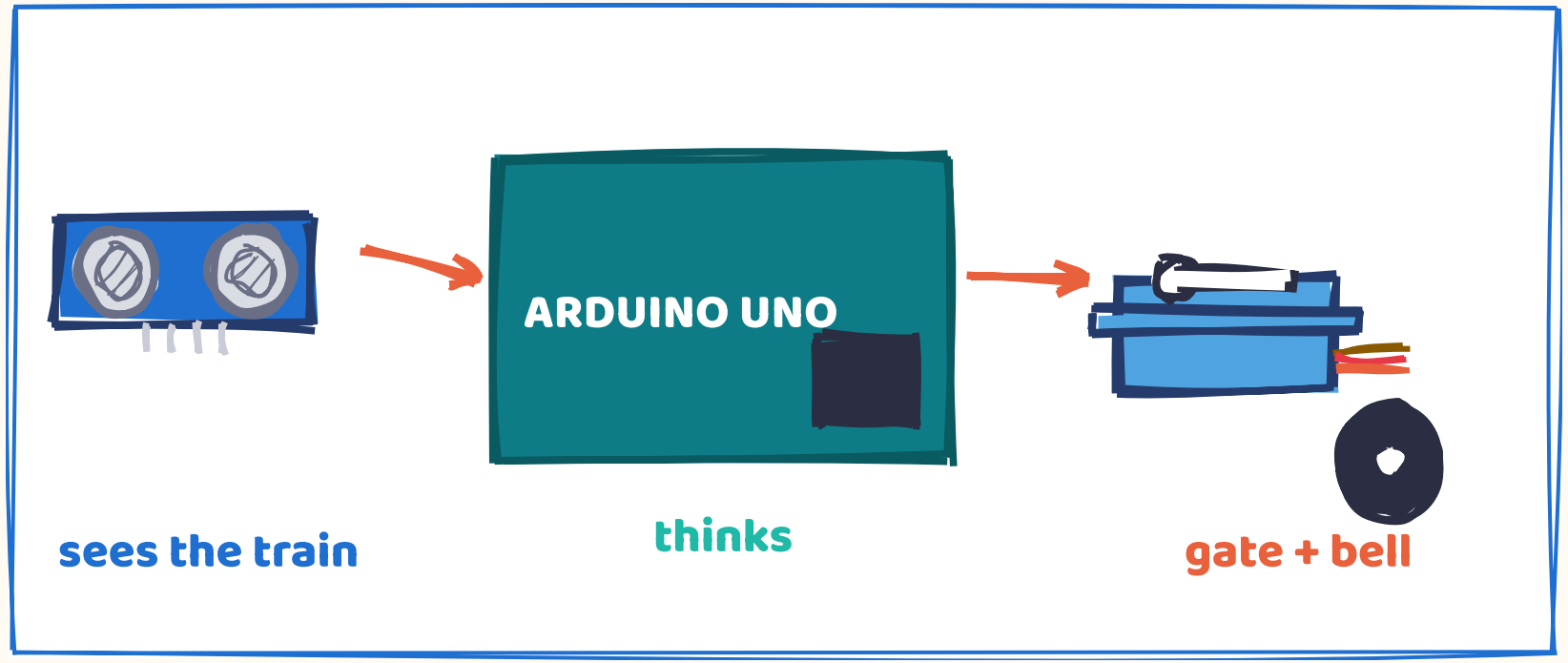
HC-SR04

Shout **"Hello!"** in a big empty hall or a cave. A moment later you hear **"hello!"** back. That is an echo. Bats find their way with echoes too!

LEARN

Our machine

Feel, think, show



How it works

The **sensor** measures how far the train is.
The Arduino **thinks**: is it closer than 10 cm?
Yes? The **servo** lowers the gate and the **buzzer** rings.

Teacher: parts are pre-wired on the breadboard. Children plug in the USB cable, snap the picture blocks in mBlock and test.

LEARN

Objective 2 · Move

Sense, decide, move

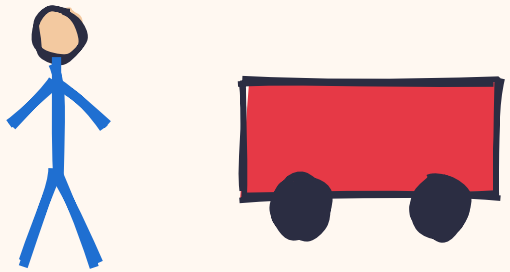
The idea

This is our first machine that does **three jobs** together:

1. **Sense**: the sensor sees the train.
2. **Decide**: the Arduino checks the distance.
3. **Move**: the servo lowers the gate.

Big idea
Almost every robot works this way: sense, decide, move!

Example: you crossing the road



Your eyes **see** a car (sense). Your brain says "too close!" (decide). Your legs **stop** (move). You are a sense-decide-move machine too!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Rules that keep us safe
2. Train coming!
3. Build the crossing
4. Snap the picture blocks
5. When should the gate close?
6. Crossing challenges

Railway safety

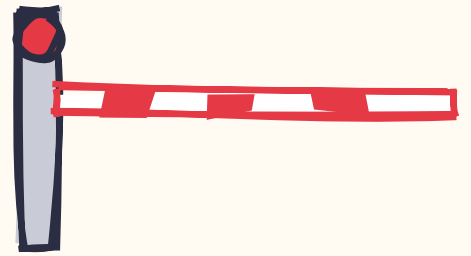
Rules that keep us safe

Stop when the lights flash and the bell rings.

Look both ways. Trains are fast and cannot stop quickly.

Never go under or around a closed gate.

Hold hands with a grown-up near a railway.



Remember

A train can take a whole kilometre to stop. It cannot swerve!

Class game

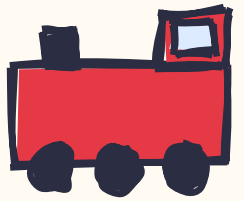
Train coming!

How to play

1. Children line up as a **train** and choo-choo around the room.
2. One child is the **gate** with an arm up.
3. When the train gets close, the gate says **"Ding ding!"** and lowers the arm.
4. Everyone else must **freeze** at the gate.
5. Train gone? Arm up, everyone crosses!

Then discuss

How close should the train be before the gate goes down? Too late is dangerous!



Big idea

Safety machines act EARLY, before the danger arrives.

Objective 2 · Build step 1 of 2

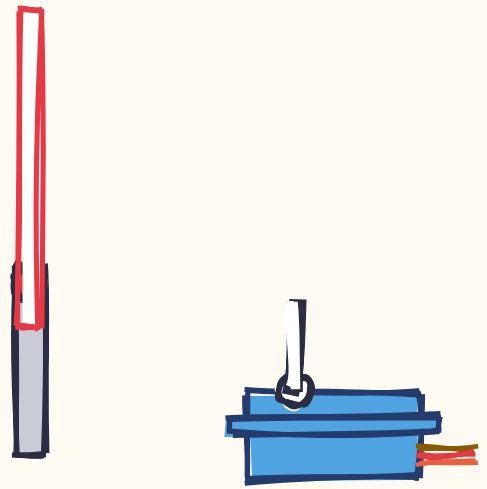
Build the crossing

a Tape **two ice cream sticks** together to make a long gate arm.

b Stick the arm onto the **servo horn**.

c Put the distance sensor **beside the track**, facing the train.

d Add the **red lights** on a card post next to the gate.



Check: The gate arm can move from up (90°) to down (0°) without hitting anything.

MAKE

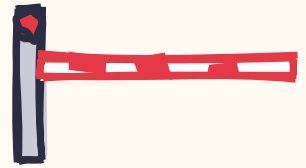
Time to build!

Snap the picture blocks

```
when Arduino starts
forever
    set distance to ultrasonic (pins 4, 5)
    if distance < 10 cm
        set gate servo (pin 9) to 0 (down)
        flash red lights + ring bell
    else
        set gate servo to 90 (up)
        lights off, bell off
```

Read it like a story

Measure how far the train is.
IF it is closer than 10 cm: gate down, lights, bell!
ELSE: gate up, all quiet.



10 cm?
Try 5, 10 and 20. Which gives cars the most time to stop?

Experiment!

When should the gate close?

Train distance	I think…	What happened
5 cm	too late / just right / too early?	_____
10 cm	too late / just right / too early?	_____
20 cm	too late / just right / too early?	_____

Talk about it

Why is closing **a bit early** safer than closing **a bit late**?

Push slowly

Push the train slowly towards the sensor. Watch when the gate moves.

Level up!

Crossing challenges

Choose your challenge



Flashing lights: make the two red lights blink one after the other.



Ding ding: make the bell ring in a ding-ding pattern.



Slow gate: lower the gate slowly, a little at a time.



Two tracks: add a second sensor for trains from the other side.

Think like an engineer

What should happen if a car is stuck on the track?

Boss level

Keep the gate down for 3 seconds AFTER the train has gone.

Objective check

Can I really do it? Prove it.

01 How does the sensor know the train is close? Use the word **echo**.

02 Show the gate going down when the train comes.

03 Tell your partner one railway safety rule.

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Run a whole-class 'train test': every group's gate must close for the same toy train.

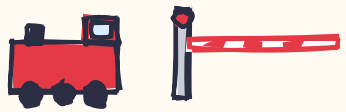
Show what you know

What did we learn?

- [x] The distance sensor uses echoes
- [x] Close train = gate down + bell
- [x] Sense, decide, move: a real robot system
- [x] Always stop at a closed crossing

At home

Next time you see a railway crossing, watch it carefully. What happens first?



Exit ticket: what three jobs does our crossing do?

Next class: build a Vibrating Brush Robot!